

Jonathon "Cole" Turner

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PROFESSIONAL SUMMARY

Full-stack software engineer and embedded systems developer with two years of professional experience building production systems at a federal contractor. Proficient in Python, Java, C, C++, React, SQL, and AWS. Built and shipped a full-stack web application with AWS cloud infrastructure and an autonomous robotics system with CNN-based real-time classification. Comfortable across the full stack and experienced leading Agile teams on systems where correctness and reliability matter.

EDUCATION

Embry-Riddle Aeronautical University

May 2026

Bachelor of Science, Computer Science. Honor Roll, GPA: 3.3

WORK EXPERIENCE

Metas Solutions – Software & IT Intern

May 2023 – June 2025

- Designed and built a full-stack PowerApps and Power BI application backed by a relational Azure database, replacing a manual process with a validated system of record for 200+ employee assets; eliminated over a week of annual reconciliation work and improved reporting accuracy for federal contract assignments
- Automated quarterly email workflows to collect and sync CDC compliance data across hardware inventory, contract assignments, and employee credentials, maintaining data integrity for sensitive federal health program records across multiple dependent systems
- As a federal consultant, led structured requirements gathering sessions with department leads for a CDC-commissioned white paper for CDC NCIRD's Vaccine for Children's Program; synthesized compliance findings into formal documentation delivered to the client.

PROJECTS

Autonomous Wildlife Deterrence System (BlackDog Enforcer)

Aug 2025 – May 2026

Senior Capstone | Led 4-person Agile team

- Developed a C++ and Python autonomous navigation engine processing LiDAR spatial data in real time using hash-map-accelerated spatial indexing; designed a dual Raspberry Pi async Wi-Fi communication protocol handling concurrent sensor streams across the perception and navigation subsystems
- Trained and deployed a CNN model for real-time wildlife classification using ZED stereo camera data; managed the full ML pipeline from labeled dataset collection through model training, validation, and production deployment on constrained embedded hardware with real inference-speed and memory constraints
- Led a 4-person Agile team from requirements through field deployment; authored System Requirements Specification and System Design documents; isolated a root-cause power degradation failure mode through structured subsystem testing, enabling safe operational deployment of the rebuilt RC vehicle platform

College Baseball Dashboard

2026

Full-Stack Web Development / AI-Assisted Engineering

- Built a full-stack React and TypeScript web application consuming ESPN REST APIs with live state polling every 20 seconds; designed a typed data integration layer, clean component architecture, and provisioned AWS CDK infrastructure as code with AWS Amplify authentication
- Implemented an SVG-based live Gamecast visualization rendering real-time play-by-play data with custom animation states; applied WCAG accessibility patterns throughout, including focus trapping, aria-labels, and prefers-reduced-motion support across the full component library
- Built a complete design token system for consistent visual language across the application; used Claude Code CLI and the Stitch MCP server for AI-assisted development, establishing CI/CD practices and a reusable component library deployable from a single infrastructure command

ATHLETICS

Collegiate Baseball

Aug 2021 – May 2026

- Academic All District 2026; Sunshine State Conference Honor Roll 2025 and 2026
- Competed for nationally ranked Division II programs; advanced to NCAA Super Regionals (2023); balanced approximately 30 hours per week of training and travel with full-time coursework throughout collegiate career

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, TypeScript, JavaScript, C++, Shell Scripting, Assembly (ARM)

Frontend: React, Tailwind CSS, Framer Motion, Vite, SVG

Backend & Cloud: RESTful APIs, AWS (Amplify, CDK), Azure, PowerApps, Power BI

Embedded & Hardware: Raspberry Pi, Arduino, STM32, GPIO/UART/SPI/I2C, PWM, OpenCV, ZED SDK

Tools & Practices: Git/GitHub, Linux/Unix, Claude Code, GDB, Agile/Scrum, CI/CD, ML

Specialized: AES/RSA Cryptography, CNN Deployment, Real-Time Systems, Data Pipelines